SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
IILM UNIVERSITY, GREATER NOIDA

UI/UX DESIGN DOCUMENT

Digital Twin of a University Building for Occupancy-Aware Energy and Indoor Environmental Monitoring

CampusTwin EnergyMind

Item	Value
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Primary Implementation Baseline	SRS v1.0
Guide	Mrs. Bhavana Srivastava, Assistant Professor

SOURCE BASIS PRD, SRS, System Architecture Document (SAD), and Phase-I Major Project Report. SRS is primary for UI requirements, permissions and acceptance; SAD governs implementation boundaries; PRD governs product goals; Phase-I informs the Digital Twin vision and future technology direction.

Document Control & Source Alignment

Source	Role in this UI/UX document	Key UI implications
SRS v1.0	Primary implementation and UI baseline	Roles, permissions, required screens, states, acceptance criteria, 3-building room mapping.
SAD v1.0	Technical implementation boundary	React frontend, API-mediated data access, role enforcement at backend, representation-neutral spatial layer.
PRD	Product/feature baseline	2D floor map MVP, occupancy/energy visibility, waste alerts, analytics and explicit non-goals.
Phase-I Report	Concept/research baseline	3D Digital Twin vision, indoor environmental monitoring, ESP32/MQTT and future analysis direction.

SCOPE STATUS The PRD explicitly describes a lightweight 2D MVP and excludes complex 3D/WebGL/VR, while the Phase-I report describes a 3D Digital Twin. The SRS and SAD retain the 2D-vs-3D direction as TBD. This document therefore defines a representation-neutral spatial UI: the current layout is implementable as 2D SVG/JSON and can later host a 3D renderer without changing navigation or room-detail structure.

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1. INTRODUCTION

1.1 Purpose

This document defines how users experience and interact with CampusTwin EnergyMind. It translates the source requirements into an implementation-ready interface structure covering navigation, screen layouts, information hierarchy, interaction states, data visualization, accessibility, responsive behavior and role-aware presentation. It is a UI/UX baseline, not a replacement for the SRS or System Architecture Document.

1.2 Product Overview

CampusTwin EnergyMind is a web-based monitoring and decision-support prototype that associates room-level occupancy, indoor environmental conditions, energy consumption, HVAC state and lighting state with a spatial representation of university building spaces. The interface supports current-state visibility, historical analysis, energy-waste detection, alerts and evidence-linked recommendations. The architecture deliberately stops at decision support and does not issue physical actuator commands.

1.3 UI/UX Goals

- Make building and room state understandable within a few seconds.
- Prioritize current occupancy, energy/resource state and active waste alerts.
- Move users from awareness to room-level evidence without unnecessary navigation.
- Make missing or stale telemetry explicit instead of showing misleading zero values.
- Keep controls role-aware and consistent with the SRS permission matrix.
- Use reusable React-friendly patterns rather than screen-specific one-off widgets.
- Support desktop-first operation at 1920×1080 and 1366×768.
- Keep the spatial presentation independent from the unresolved 2D/3D renderer decision.

> ***UI naming clarification: For this UI/UX baseline, the display label for SVH is "Swami Vivekand Hall" as instructed for the current project presentation. The source SRS v1.0 uses "Sir Vishveshwarya Hall"; this naming discrepancy does not change the mapped room IDs or floor structure.***

1.4 Target Users

Role	Primary UI needs
Facility / Building Administrator	Building-wide dashboard, utilization trends, alerts/anomalies, administrative functions permitted by SRS.
Energy / Resource Manager	Occupancy-energy correlation, waste alerts, historical analysis, recommendations.
Campus Operations Staff	Room/resource status, operational waste locations, inspection-oriented alerts.
Faculty / Student	Read-only room availability and current occupancy; other metrics remain role-dependent/proposed.

1.5 UI/UX Scope

Status	Included
Confirmed / source-supported	Login/logout, RBAC-aware navigation, three-building spatial navigation, room details, occupancy, temperature, humidity, CO_2, energy, HVAC, lighting, alerts, recommendations, 24h/7d/30d analytics, loading/empty/error states.
Proposed / implementation choice	Exact visual theme, icon library, chart library, session duration, some operational alert lifecycle details.
TBD / unresolved	Final 2D versus 3D spatial renderer; exact retention periods and certain thresholds where not specified.
Future scope	Additional buildings/sensors, occupancy prediction, predictive maintenance, BMS integration, campus federation, native mobile apps, VR/AR.

2. UX DESIGN PRINCIPLES

Clarity First

Show the current building/floor/room context at all times. Use plain labels such as Occupied, Unoccupied, Available and Data unavailable.

Real-Time Awareness

Give current state visual priority and show a freshness timestamp so users can distinguish current data from stale data.

Actionable Information

An alert should explain the room, issue, duration and available wasted-energy evidence; a recommendation should show the evidence behind the suggested human action.

Minimal Cognitive Load

Use a stable layout, limited status colors, short labels and progressive disclosure for room-level detail.

Consistent Interaction

Building selection, floor selection, filters, room selection and status cards behave consistently across screens.

Evidence-Based Recommendations

Recommendation cards connect issue → supporting metric → suggested human action → severity → timestamp/status.

Safe Decision Support

The interface must never imply that a recommendation is an actuator command. Any operational action is performed by authorized people outside the system.

Accessibility

Status must not depend on color alone; text, iconography, labels and accessible states are required.

3. USER PERSONAS

Persona	Context	Key goals	Typical UI path
Facility / Building Administrator	Monitors overall building efficiency and anomalies.	See load, occupancy, alerts, utilization trends and permitted administrative controls.	Login → Dashboard → Building/Floor → Alerts/Analytics → Room evidence.
Energy / Resource Manager	Investigates energy consumption and waste patterns.	Compare occupancy with energy, inspect waste alerts and recommendations.	Dashboard → Alert → Room Detail → Analytics → Recommendation.
Campus Operations Staff	Uses system to prioritize physical checks.	Find active waste locations and inspect resource state.	Dashboard → Alerts → Room Detail → Recommendation.
Faculty / Student	Needs transparent room availability.	Find a currently available room without administrative controls.	Login if enabled → Building → Floor → Room availability.

4. INFORMATION ARCHITECTURE

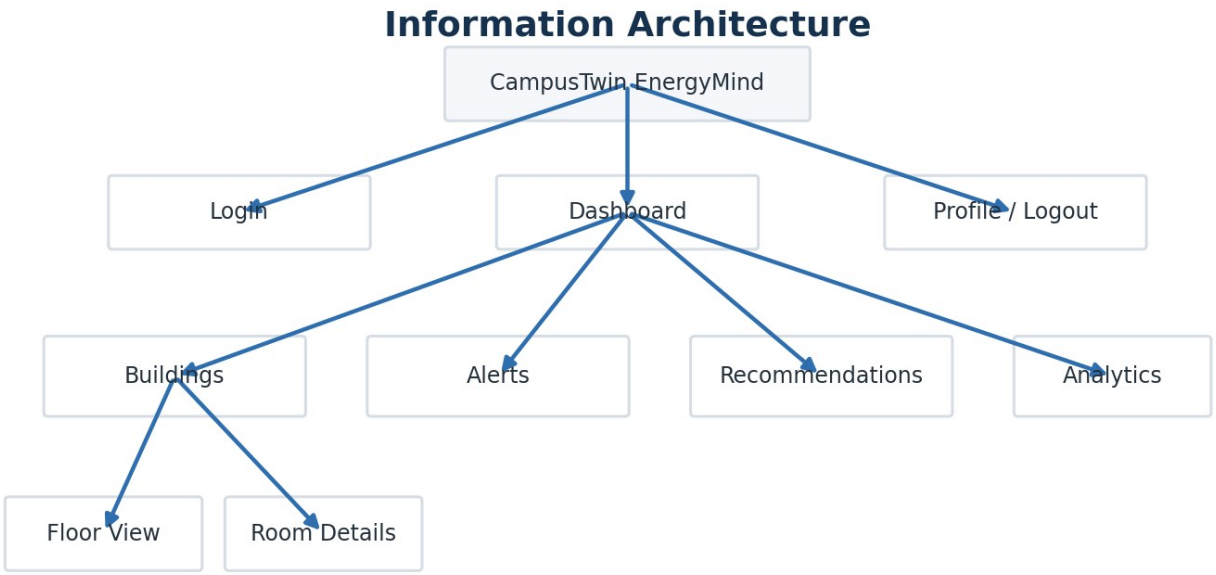


Figure 1. Information architecture. The hierarchy is intentionally shallow so users can move from building context to room evidence quickly.

Primary hierarchy: Login → Dashboard → Buildings/Floor/Room → Alerts / Recommendations / Historical Analytics. Profile/logout is a utility area. Role-based rendering changes what is visible and actionable, but does not create separate application architectures for each role.

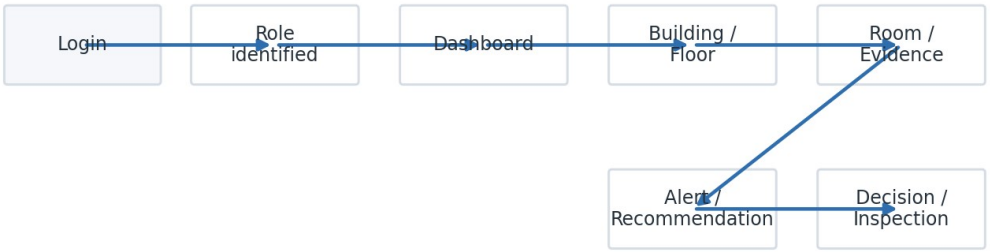
4.1 MVP Spatial Inventory

Building	Floor	Room count	Inclusive room ID range
SVH — Swami Vivekand Hall	Ground	12	101–112
SVH — Swami Vivekand Hall	Second	13	201–213
Engineering Block	Ground	8	101–108
Engineering Block	First	8	201–208
Engineering Block	Second	8	301–308
Engineering Block	Third	8	401–408
Law Block	Ground	8	101–108
Law Block	First	8	201–208
Law Block	Second	8	301–308
Law Block	Third	8	401–408
TOTAL	Current MVP	89	No additional rooms defined

ROOM RANGE CONVENTION Room ranges are inclusive. For example, 101–108 means eight distinct rooms: 101, 102, 103, 104, 105, 106, 107 and 108. The same convention applies to every supplied range. No unlisted rooms are introduced.

5. USER JOURNEYS

Main User Journey: Awareness → Investigation → Evidence → Decision Support



5.1 Journey 1 — Login → Dashboard

35. User opens the application and sees the login form.
36. User submits identifier and password.
37. Frontend shows loading feedback and disables repeated submission.
38. Backend validates credentials and establishes an authenticated session.
39. Role context is applied; the permitted dashboard is displayed.
40. Invalid credentials produce a generic error without revealing whether an account exists.
41. Expired sessions redirect/reject protected access and require re-authentication.

5.2 Journey 2 — Building → Floor → Room

42. Select one of the three configured buildings.
43. Select a configured floor.
44. Load the spatial representation and exact configured room IDs.
45. Select a room node/card.
46. Open the room detail panel with all valid current telemetry and timestamps.
47. If a metric is unavailable, show an explicit unavailable state rather than zero.

5.3 Journey 3 — Detect Energy Waste

48. Dashboard exposes an active waste alert.
49. User opens the alert and sees room, issue, duration, severity and available wasted kWh.
50. User opens the room detail view to inspect occupancy, HVAC/lighting, energy and environment.
51. Recommendation card presents supporting evidence and a human-operational suggestion.
52. Operations staff or another authorized user performs any physical inspection/action outside the application.

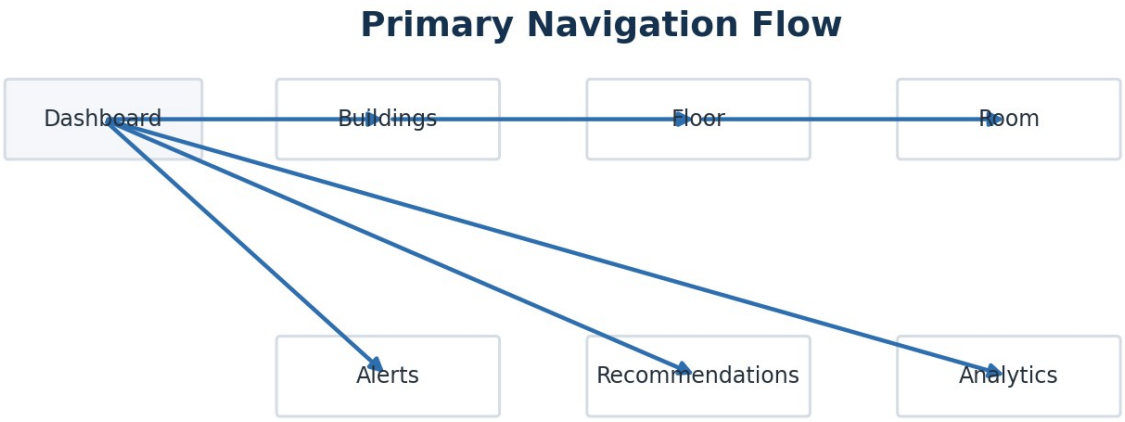
5.4 Journey 4 — Historical Analysis

Select 24 hours, 7 days or 30 days; optionally filter building/floor/room; render occupancy-versus-energy charts and supported environmental trends; show incomplete-data segments explicitly.

5.5 Journey 5 — Faculty/Student Room Availability

Where login/read-only access is enabled, the user selects building and floor and views room availability/current occupancy. No administrative, alert lifecycle, configuration or physical-control actions are presented.

6. NAVIGATION DESIGN



Navigation item	Purpose	Roles	Destination
Dashboard	Operational overview	Admin, Energy, Operations; read-only subset	Main Dashboard
Buildings	Select building and floor	All	Building / Floor View
Alerts	Review waste events	Admin, Energy, Operations	Alerts Page
Recommendations	Review evidence-linked suggestions	Admin, Energy, Operations	Recommendations Page
Analytics	Historical occupancy/energy analysis	Admin, Energy; Operations proposed	Historical Analytics
Profile / Logout	Session utility	All authenticated users	Profile / Logout

6.1 Navigation Rules

- Use a left sidebar on desktop with a persistent top header.
- Show current building/floor context in the page header or breadcrumb when inside spatial views.
- Do not show unauthorized actions; backend authorization remains authoritative.
- Use breadcrumbs only for spatial depth: Dashboard → Building → Floor → Room.
- Keep building and floor selectors close to the spatial view, not buried in a settings page.
- Use active navigation state and a visible focus state for keyboard users.

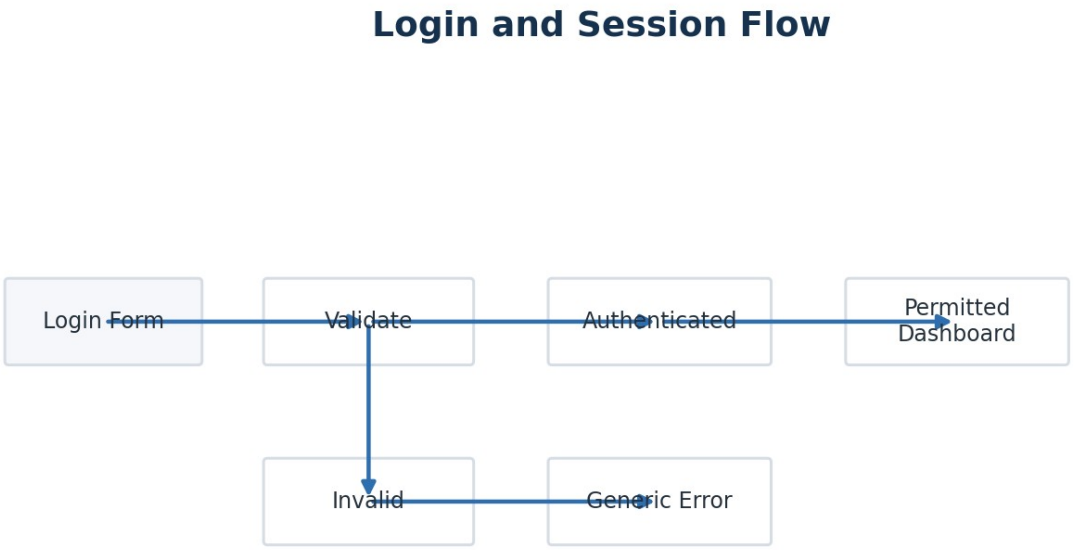
7. SCREEN INVENTORY

#	Screen	Purpose	Core status
1	Login Page	Authenticate users	Confirmed
2	Main Dashboard	High-level operational view	Confirmed
3	Digital Twin / Building View	Spatial building/floor/room visibility	Confirmed structure; 2D/3D renderer TBD
4	Floor Navigation	Move among mapped floors	Confirmed
5	Room Detail Panel/Page	Room telemetry and recent state	Confirmed
6	Alerts Page	Energy-waste events	Confirmed
7	Recommendations Page	Decision-support	Confirmed

		suggestions	
8	Historical Analytics	24h/7d/30d occupancy-energy comparison	Confirmed
9	Profile / Logout	Session utility	Utility function; no separate rich profile feature required

SCREEN MINIMIZATION No extra management screens are added. Simulator configuration, rule configuration and user/role administration are role-protected capabilities from the SRS, but their exact screen designs are implementation-dependent/proposed and are not treated as required MVP navigation destinations.

8. LOGIN PAGE UI



Layout

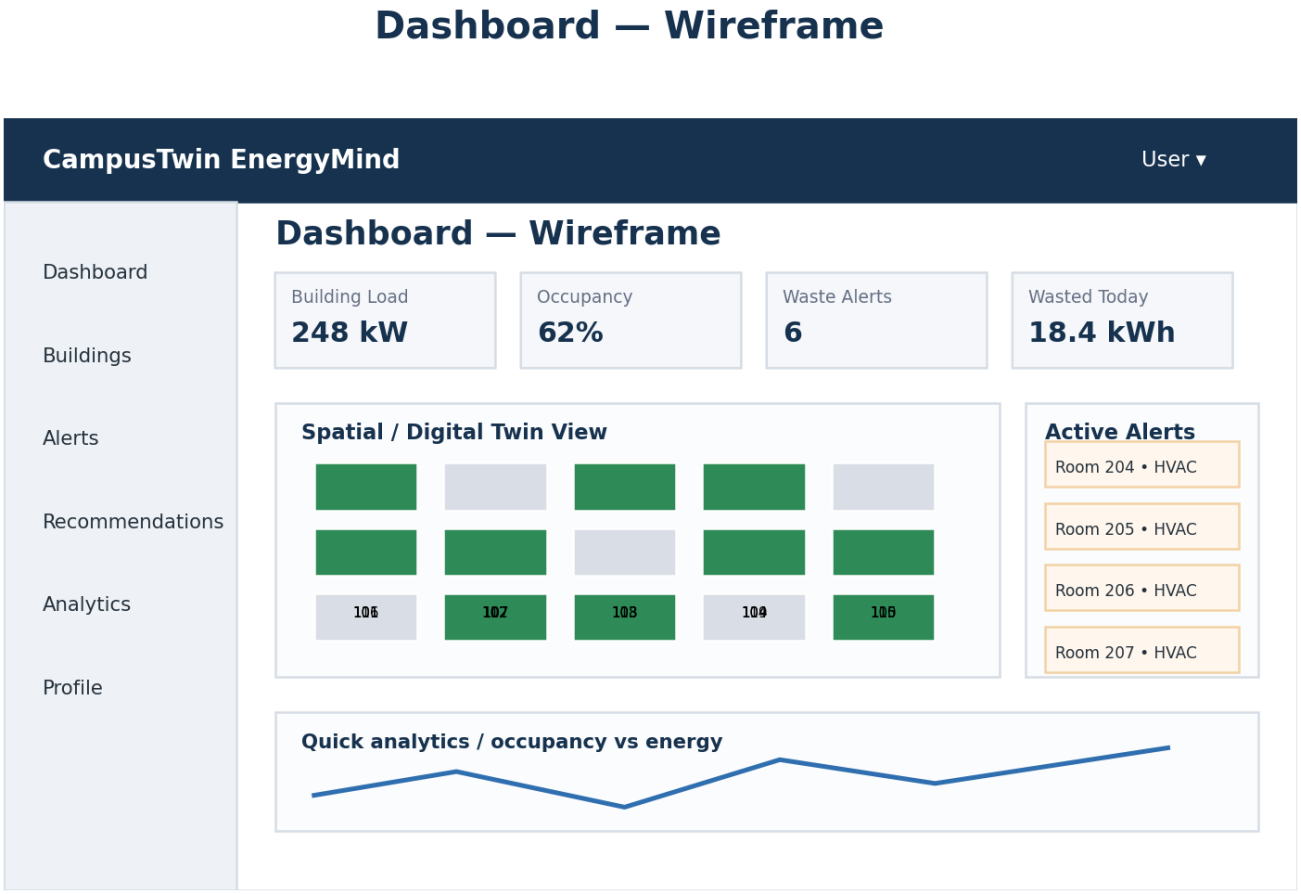
- Centered authentication card with project/university branding.
- Short product descriptor: occupancy-aware building monitoring and decision support.
- Identifier field, password field, sign-in button and generic feedback region.
- Avoid decorative background elements that reduce form contrast or distract from authentication.

Components and states

Component	Default	Interaction / feedback	Accessibility
Identifier field	Empty	Required validation; visible focus	Explicit label and autocomplete appropriate to implementation
Password field	Masked	Required validation; generic failure	Label; do not expose credentials
Sign-in button	Enabled	Loading → disabled while submitting	Keyboard accessible
Error region	Hidden	Generic authentication failure	ARIA live region where appropriate
Session expiry	N/A	Re-authentication required	Clear message + login path

SECURITY UI RULE Authentication errors must remain generic and must not reveal whether a particular account exists. Protected pages/actions require authenticated context; hiding a control is not the only security mechanism.

9. MAIN DASHBOARD UI



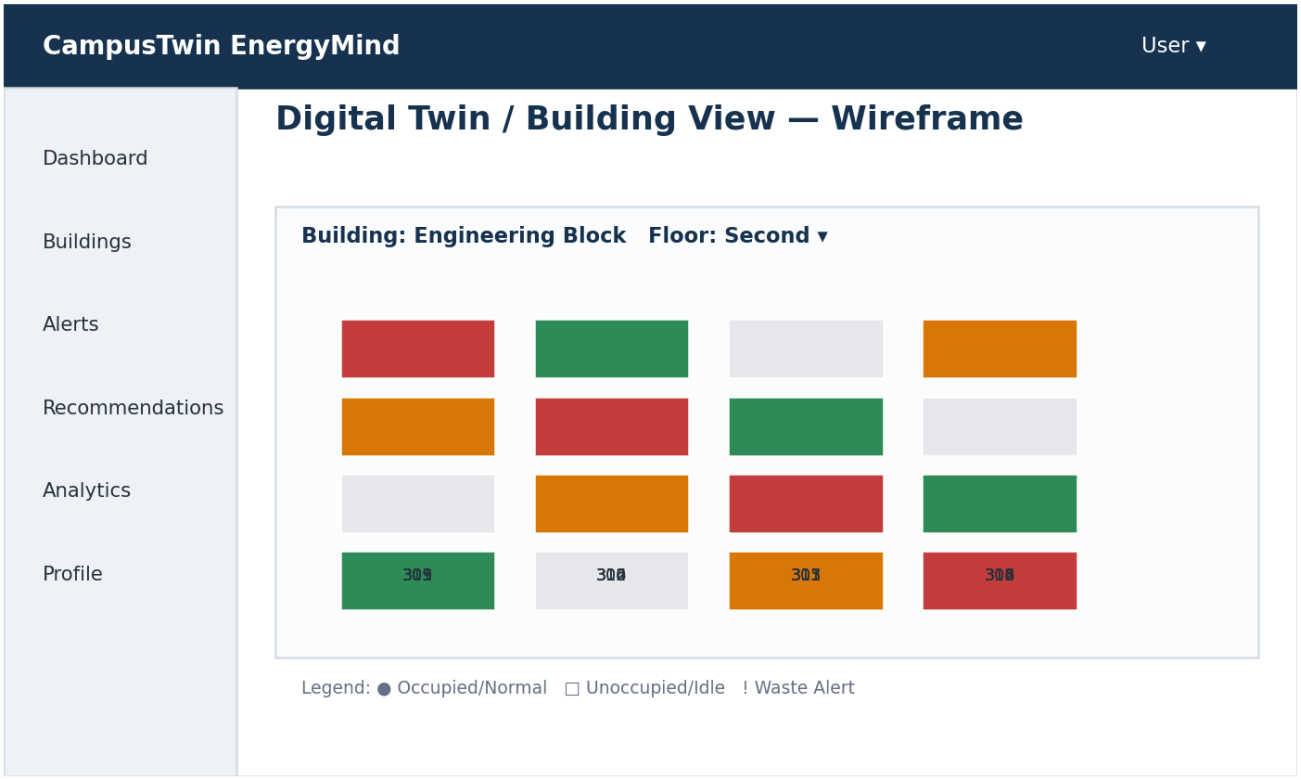
Exact component hierarchy

53. Global header: project identity, current user/role and logout.
54. Sidebar: Dashboard, Buildings, Alerts, Recommendations, Analytics.
55. Page header: selected building/floor context and last-refresh indicator.
56. Summary cards: Total Building Load, Current Building Occupancy %, Active Waste Alerts, Energy Wasted Today; System Health may be shown where supported by available data.
57. Main spatial panel: Digital Twin / floor representation with status legend.
58. Active alerts panel: room, issue, severity, duration and available wasted kWh.
59. Quick analytics: compact occupancy-versus-energy preview and link to Historical Analytics.

Dashboard behavior

- Filters update the spatial view and summary context without navigating away.
- Clicking a room opens room detail; clicking an alert opens alert evidence; analytics opens the full historical screen.
- Current-state changes should become visible within the SRS target of 2 seconds after state-change ingestion under representative load.
- Skeleton loaders preserve layout during data retrieval.
- No-data and API error states are localized to the affected widget when possible.

10. DIGITAL TWIN / BUILDING VIEW



Spatial interaction model

- Building selector lists exactly SVH, Engineering Block and Law Block.
- Floor selector lists only configured floors for the selected building.
- Room nodes preserve supplied IDs exactly.
- Room state is communicated with a color plus text/icon/badge so color is not the sole signal.
- Selecting a room highlights it and opens a detail panel; hover may provide a compact tooltip on pointer devices but all information must be reachable without hover.
- Zoom/pan is optional for 2D layouts and should be used only when it improves room selection.

2D/3D compatibility

The spatial shell consists of building selector + floor selector + spatial canvas + legend + selected-room state. In the current source baseline, lightweight SVG/JSON layouts are supported for MVP implementation. If the project team later confirms a 3D renderer, the same shell can host a Three.js/React Three Fiber view without changing room detail, alert, recommendation or analytics navigation.

Status vocabulary

State	Visual treatment	Text label
Occupied / Normal	Green accent + person/occupancy icon	Occupied
Unoccupied / Idle	Neutral gray accent + empty-room icon	Unoccupied
Available	Neutral/positive badge	Available
Waste Alert	Red accent + warning icon/pattern	Waste alert
Data unavailable	Dashed/neutral treatment	Data unavailable
Stale telemetry	Amber accent + clock icon	Stale

11. FLOOR NAVIGATION

State	UI behavior
Loading	Keep building context visible; skeleton the floor selector/map region.
Successful load	Show current floor label, room count and configured room nodes.
No configured floor	Show clear empty state: “No configured floor available.”
Invalid floor	Show controlled error and return to building selection.
Data unavailable	Keep spatial structure if known and mark room telemetry unavailable.

12. ROOM DETAIL UI



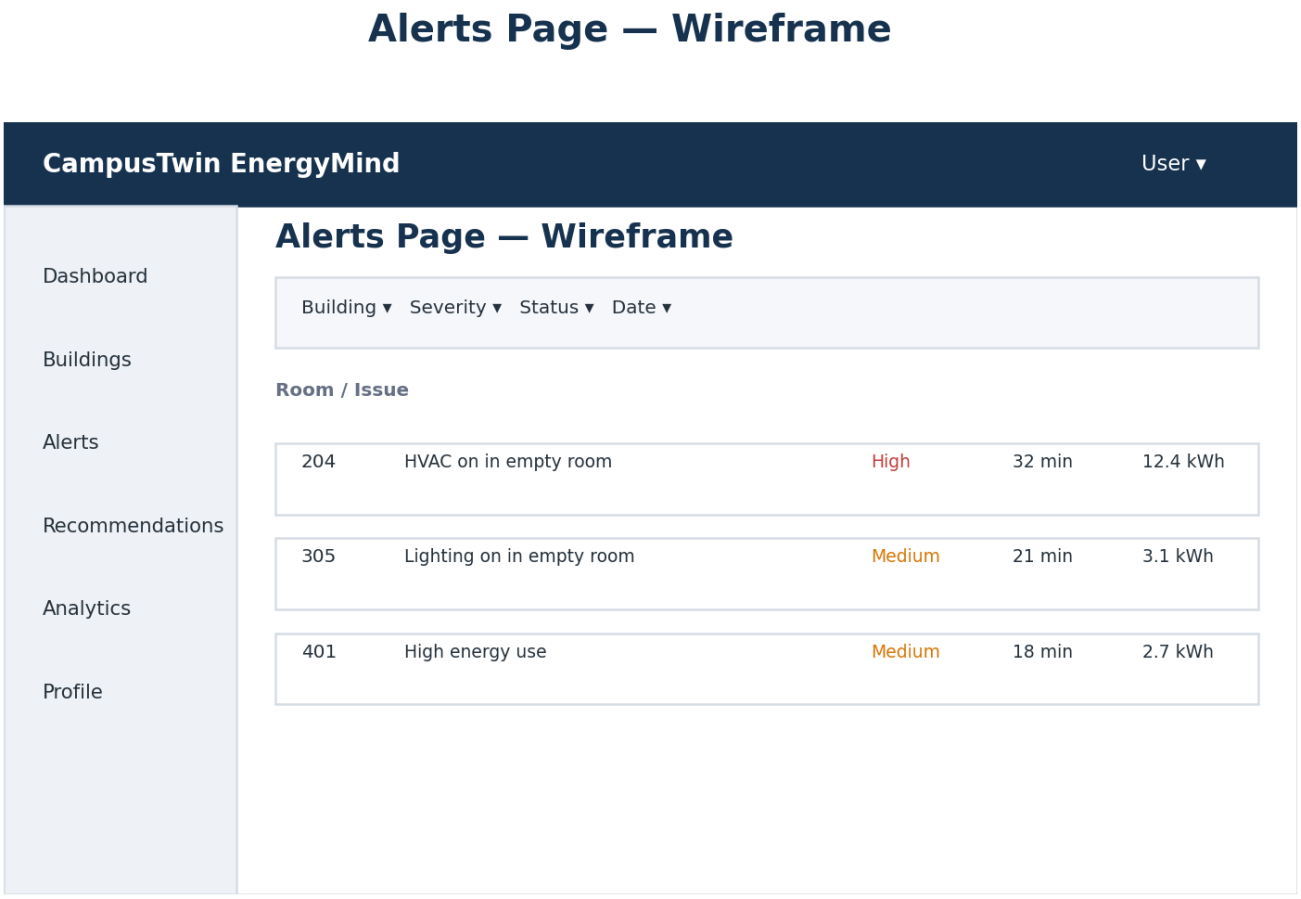
Room identity and telemetry

Group	Required display
Identity	Room ID; room type/capacity only if known/configured.
Occupancy	Occupied / Unoccupied / Available; count when available; timestamp.
Environment	Temperature, humidity, CO_2; each with value, unit and timestamp when valid.
Energy	Active power and/or energy consumed with units and timestamp.
Resources	HVAC: Active/Idle/OFF; Lighting: ON/OFF/Dimmed.
History	24-hour preview where available.
Related information	Linked alerts and recommendations.

DATA INTEGRITY If a metric is missing, invalid, stale or unsupported, display an explicit state such as

"Data unavailable" or "Last updated …". Never infer zero from missing telemetry.

13. ALERTS PAGE



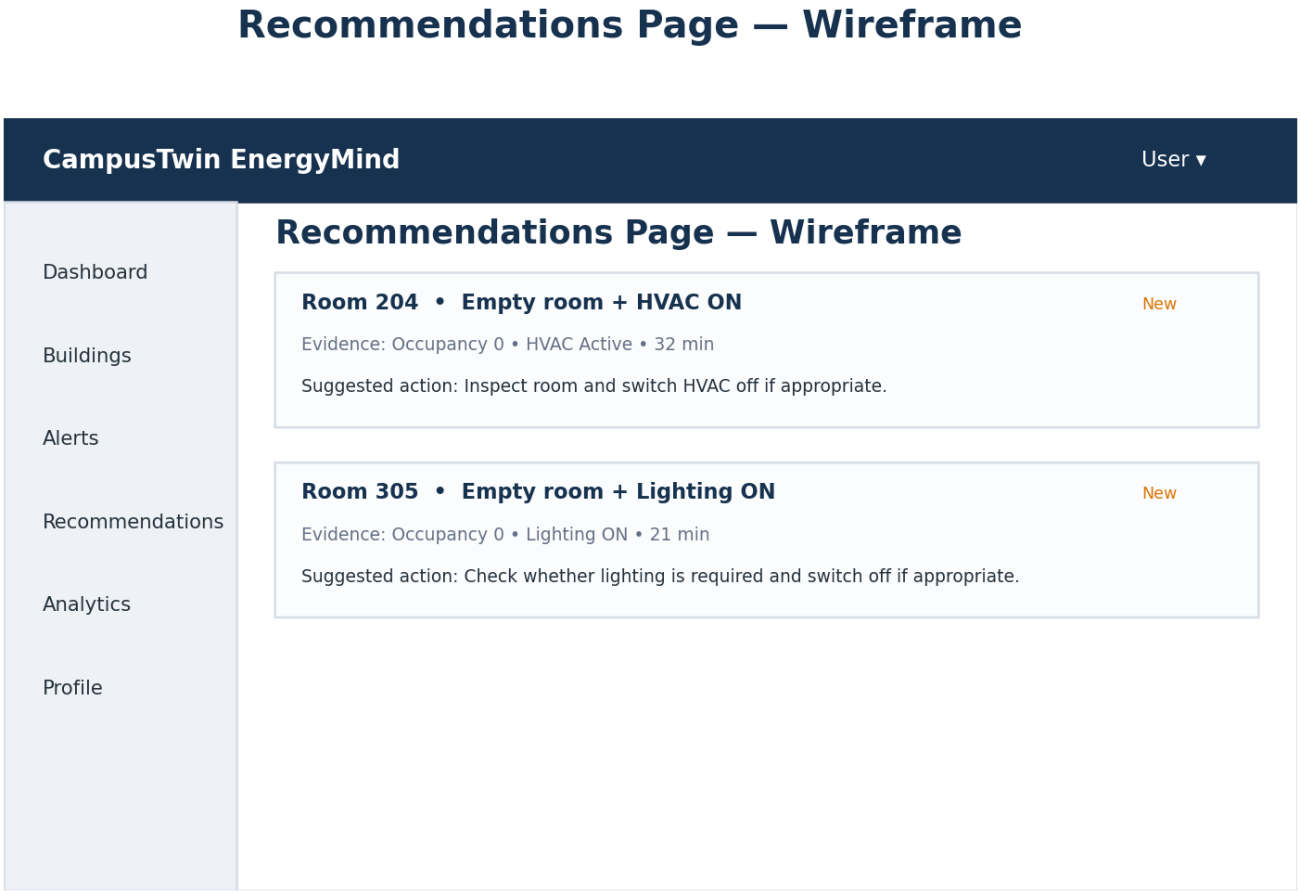
Alert structure

Field	UI treatment
Room	Clickable room ID / context link
Issue type	Short operational phrase such as HVAC on in empty room
Trigger time	Timestamp
Duration	Human-readable duration; exact timestamp available on detail
Severity	Text badge + icon; color is secondary
Status	New/Acknowledged/Resolved only if lifecycle is configured
Estimated wasted kWh	Value when sufficient data exists; otherwise unavailable
Actions	Filter/open/acknowledge/resolve only when permission/configuration allows

Alert states

- Loading: skeleton rows/cards.
- No alerts: positive neutral empty state, e.g. "No active energy-waste alerts."
- Error: explain that alerts could not be loaded and offer Retry.
- Unauthorized: hide the module or show a controlled forbidden state according to role handling.

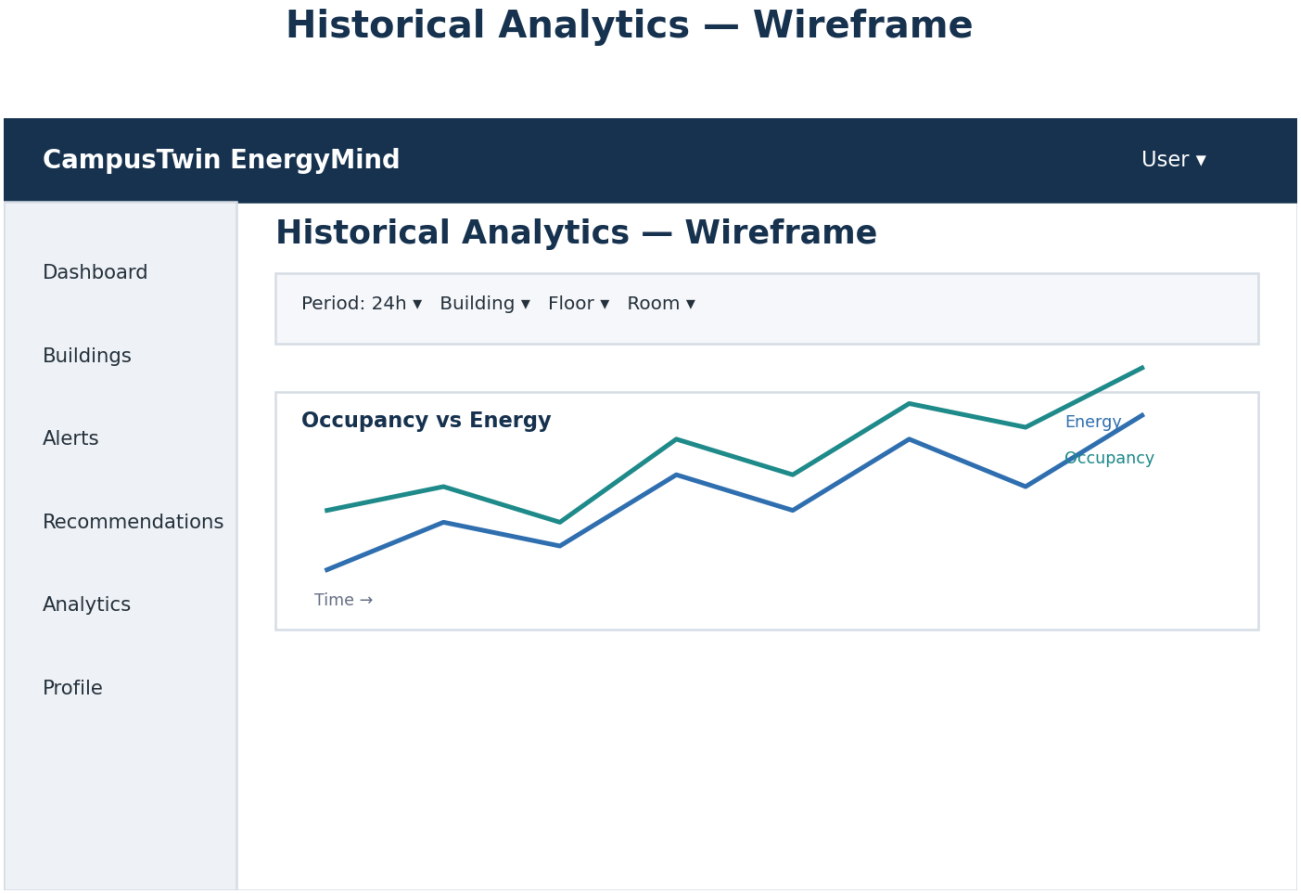
14. RECOMMENDATIONS PAGE



Recommendations are evidence-linked decision-support outputs. They never become direct actuator commands. The SRS requires issue, room, supporting metric, suggested action, severity, timestamp and status.

Recommendation trigger	Evidence shown	Human-oriented action pattern
Empty room + HVAC ON	Occupancy = 0; HVAC state; duration; power/energy	Inspect room and switch HVAC off if operationally appropriate.
Empty room + Lighting ON	Occupancy = 0; lighting state; duration	Check whether lighting is required and switch off if appropriate.
High occupancy	Occupant count and capacity where configured	Review capacity/utilization and operational alternatives.
High CO_2	CO_2 value + timestamp	Inspect ventilation/air-quality conditions.
Abnormal temperature	Temperature + trend	Review HVAC/environmental conditions.
High energy consumption	Active power/energy + occupancy/baseline context	Inspect resource usage and compare with context.
Prolonged resource usage	Duration + resource state	Prioritize room for physical inspection.

15. HISTORICAL ANALYTICS UI



Required controls

- Period selector: 24 hours, 7 days, 30 days; custom range only if supported by the implemented API.
- Filters: building, floor and room.
- Chart toggle/selection should remain simple; avoid excessive dashboard controls.

Required visualizations

Visualization	Recommended form	Units / labels
Occupancy vs Energy	Dual-series line chart or aligned time-series view	Occupancy count/%; kW or kWh
Hourly/daily occupancy	Line or bar chart	Occupants or occupancy %
Energy consumption	Line/bar chart	kW/kWh
Environmental trends	Line chart per supported parameter	°C, %, ppm
Room utilization heatmap	Grid/heatmap where supported	Room × time; utilization scale

PERFORMANCE TARGET The SRS specifies a client-visible 30-day historical chart rendering target of no more than 1.5 seconds for the documented dataset target. Use aggregated API responses and avoid loading unnecessary raw telemetry into the browser.

16. COMPONENT LIBRARY

Component	Purpose	Variants / states	Accessibility
Sidebar	Primary navigation	Active, inactive, collapsed if needed	Keyboard focus; semantic nav
Header	Global context	Default, session-expiry warning	Landmarks; readable labels
Breadcrumb	Spatial context	Dashboard/Building/ Floor/Room	Text-based navigation
Building selector	Choose building	Default, open, loading, invalid	Label + keyboard
Floor selector	Choose floor	Configured, loading, unavailable	Label + keyboard
Metric card	Show high-level KPI	Normal, warning, unavailable	Label/value/unit; no color-only status
Status badge	Compact state	Occupied, Unoccupied, Available, Alert, Stale	Text + icon
Room node/card	Select room	Default, hover, selected, alert, stale	Accessible name + selected state
Alert card/table row	Review alert	New, acknowledged, resolved, unavailable kWh	Semantic row/button
Recommendation card	Show evidence/action	New, acknowledged, resolved	Clear hierarchy; action text
Data table	Dense operational data	Loading, empty, error, pagination if needed	Header semantics + keyboard
Chart container	Host visualization	Loading, data, incomplete, error	Text summary/description
Toast/banner	Transient feedback	Info, success, warning, error	ARIA live where appropriate
Skeleton	Loading placeholder	Card, table, map, chart	Not announced as content
Empty state	No valid records	No alerts, no history, no rooms	Clear message + next action
Error state	Recoverable failure	API, network, query	Actionable Retry

16.1 Component interaction rule

Reusable components should receive state and data through React props/state rather than duplicating business logic. The visual component should render the state; data validation, permissions and business rules remain in the application/API layers.

17. DESIGN SYSTEM

17.1 Color System

Token	Suggested value	Use
Primary	Deep navy #16324F	Header, major headings, primary navigation
Secondary	Blue #2F6FB0	Links, active navigation, data emphasis
Accent	Teal #1F8A8A	Secondary emphasis, supporting analytics
Background	#F5F7FA	Application canvas
Surface	#FFFFFF	Cards, panels, tables
Text	#25313C	Primary text
Muted text	#667085	Secondary text, timestamps

Border	#D7DDE5	Panel/card/table boundaries
Success	#2E8B57	Normal/positive state
Warning	#D97706	Warning/stale state
Error	#C43D3D	Waste/error state
Info	Blue/teal	Informational feedback

Semantic status colors are secondary cues. Every status must also have a readable label, icon, pattern, or text indicator. Avoid using many unrelated accent colors.

17.2 Typography

Level	Guideline
Page title	24–28 px equivalent; bold; short.
Section heading	16–18 px; bold.
Subheading	12–14 px; semibold.
Body	14–16 px equivalent for primary content; 9–10 pt document rendering.
Metric value	24–32 px equivalent; strong numeric hierarchy.
Label/caption	11–13 px; muted but readable.

17.3 Spacing / radius / shadows

Use a 4/8/12/16/24/32 px spacing scale. Use small, consistent 6–10 px corner radii. Keep shadows subtle; borders should carry most of the panel separation. Avoid decorative depth.

17.4 Icons

Use one consistent outline/icon style for occupancy, energy, HVAC, lighting, warning and navigation. A specific icon library is Proposed because the source documents do not finalize one.

18. INTERACTION DESIGN

Trigger	System response	Visual feedback
Room click	Load/refresh room context	Selected outline + detail panel + loading skeleton if needed
Building change	Fetch configured floors and building state	Selector updates; map skeleton
Floor change	Load floor room layout/state	Current floor highlighted; room nodes appear
Filter change	Refresh visible data	Filter chip/label; loading state localized
Alert open	Load alert evidence and related room	Alert detail context + room link
Recommendation open	Show evidence and suggested action	Expanded card/detail state
Chart hover/focus	Show exact value/time	Tooltip plus keyboard-accessible equivalent
Submit login	Authenticate	Button loading/disabled; generic error if failure
Retry	Re-run failed request	Button loading; state restored on success

19. FORMS AND VALIDATION

19.1 Login Form

Field	Validation	Feedback
Username / identifier	Required	Inline required message; preserve entered value where safe.
Password	Required	Inline required message; masked input.
Submit	Both fields valid	Loading state; disable duplicate submissions.
Authentication	Credentials validated server-side	Generic failure; no account enumeration.

19.2 Other forms

No additional user-facing forms are treated as mandatory MVP UI screens. Simulator configuration, threshold configuration and user/role administration are protected capabilities in the SRS; their exact form fields and screen layouts are implementation-dependent/proposed and should be specified only when those features are enabled.

20. LOADING STATES

Area	Loading behavior
Login	Button progress state; prevent duplicate submit.
Dashboard	Skeleton cards, spatial panel and alert rows; preserve layout.
Building/Floor	Skeleton map and selector while configuration loads.
Room telemetry	Skeleton metric cards; retain room identity.
Alerts/Recommendations	Skeleton list/table rows.
Historical charts	Chart skeleton + selected period remains visible.

21. EMPTY STATES

Condition	Message pattern	Next action
No alerts	No active energy-waste alerts.	Review dashboard or refresh.
No recommendations	No recommendations are currently available.	Return to dashboard.
No telemetry	Telemetry is not available for this room.	Wait/retry or inspect data source.
No history	Historical data is unavailable for the selected period.	Change period/room if permitted.
No configured floor	No floor is configured for this building.	Return to building selection.
No room data	Room is configured but current data is unavailable.	Show room structure; do not fabricate values.

22. ERROR STATES

Error	User-facing response
Invalid login	“Sign-in failed. Check your credentials and try again.”
API/network failure	“We could not load this data. Retry.”
Floor load failure	“This floor could not be loaded. Retry or return to

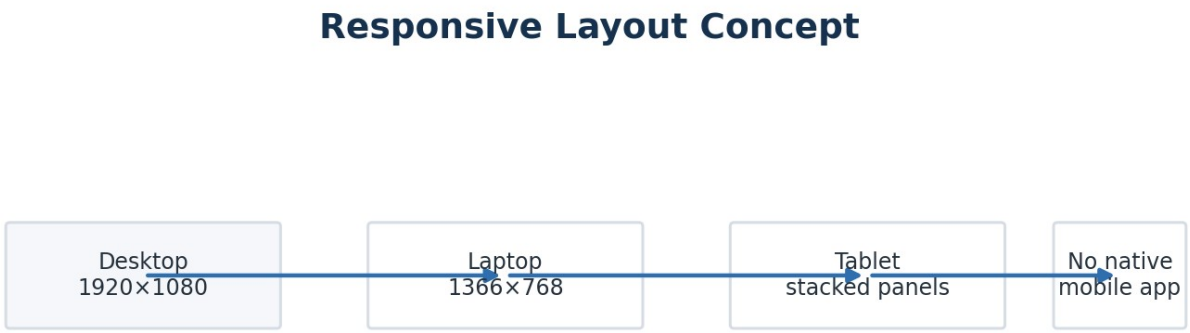
	building selection.”
Room telemetry failure	“Room data is temporarily unavailable.”
Historical query failure	“The selected history could not be loaded. Retry or change the period.”
Invalid building/floor/room	“The selected space is not available.”
Session expiry	“Your session has expired. Sign in again.”
Forbidden	“You do not have permission to access this function.”

ERROR DETAIL RULE Normal users receive structured, actionable messages rather than stack traces, database errors, credentials or implementation details.

23. DATA FRESHNESS AND STALE DATA UI

- Every current telemetry group should expose a last-updated timestamp when data exists.
- Stale telemetry should be visibly marked rather than silently presented as current.
- Offline sensors should be shown as offline/stale; do not infer zero occupancy or zero energy.
- Partial room data is valid: one metric can be unavailable while another remains visible.
- Use concise labels such as “Last updated 12 sec ago”, “Data unavailable” and “Sensor offline”.

24. RESPONSIVE DESIGN



Viewport	Behavior
1920×1080	Full sidebar, 4 KPI cards in one row, large spatial panel, alerts beside/under spatial view.
1366×768	Same hierarchy with tighter spacing; KPI cards may wrap 2×2; room detail can use a side panel or stacked panel.
Tablet / smaller browser	Collapse sidebar; stack KPI cards; spatial view remains primary; room detail becomes a lower/overlay panel.
Native mobile app	Out of scope. The web interface may adapt to smaller widths but no iOS/Android application is designed.

24.1 Responsive priorities

- Preserve room selection and current context above decorative content.
- Never hide the alert/recommendation evidence needed for decision support.

- Allow horizontal scrolling for dense tables only when necessary; prefer responsive card views at smaller widths.
- Charts should maintain readable labels and support horizontal space reduction without becoming misleading.

25. ACCESSIBILITY

- Keyboard navigation for all interactive controls.
- Visible focus states with sufficient contrast.
- Semantic HTML landmarks for header, nav, main and sections.
- Explicit form labels; do not rely on placeholder text as the only label.
- ARIA labels/live regions where needed for status updates and feedback.
- Status communicated by text/icon/pattern in addition to color.
- Chart summaries and accessible data descriptions for key trends.
- Tooltips cannot be the only way to access important information.
- Logical tab order follows visual reading order.
- Clear, actionable error messages.
- Do not display PII or individual tracking information.

ACCESSIBILITY TARGET Use a practical WCAG-aligned design approach. This document does not claim formal WCAG certification unless separately tested and approved.

26. ROLE-BASED UI BEHAVIOR

UI Feature	Facility Admin	Energy Manager	Operations Staff	Faculty / Student
Login / logout	View	View	View	Yes, if enabled
Building/floor/room status	View	View	View	Read-only
Occupancy	View	View	View	Read-only
Environmental metrics	View	View	View	Proposed read-only
Energy/resource state	View	View	View	Proposed read-only
Alerts	View	View	View	No / Proposed
Acknowledge alert	Proposed	Proposed	Proposed	No
Resolve alert	Only if configured/authorized	Only if configured/authorized	Only if configured/authorized	No
Waste-rule thresholds	Proposed	Proposed	No	No
Simulator	Yes / Proposed	Yes / Proposed	Proposed	No
Historical analytics	View	View	Proposed	No / Proposed
Export summaries	View	View	Proposed	No
User/role administration	Proposed	No	No	No

RBAC IMPLEMENTATION RULE The backend/API is authoritative. UI hiding is a usability measure, not the security boundary. Unauthorized actions must be rejected server-side.

27. UI STATE MODEL

Page	Default	Loading	Success	Empty	Partial / stale	Error	Unauthorized / session
Login	Form	Submit progress	Dashboard redirect	Required-field validation	N/A	Generic auth error	Re-authentication
Dashboard	Current context	Skeleton widgets	Live data	No-data widget	Stale badge	Retry widget	Forbidden/ redirect
Building view	Building + floor	Map skeleton	Room nodes	No configured floor	Room stale badges	Map retry	Forbidden if protected
Room detail	Room identity	Telemetry skeleton	Metrics + timestamp	Unavailable metrics	Stale/offline labels	Retry	Role-limited fields
Alerts	Current filters	Skeleton rows	Alert list	No alerts	Missing kWh marked unavailable	Retry	Forbidden/ hidden
Recommendations	Current filters	Skeleton cards	Evidence cards	No recommendations	Missing evidence flagged	Retry	Forbidden/ hidden
Analytics	24h default	Chart skeleton	Chart data	No history	Incomplete segments	Retry	Role-limited

28. DATA VISUALIZATION DESIGN

Data	Preferred visualization	UI rules
Occupancy	Line/bar trend; room status badge	Show count/% and time; status text.

Energy	Line/bar trend	Label kW vs kWh correctly; distinguish instantaneous power from accumulated energy.
Temperature	Line trend	°C and timestamp; mark missing periods.
Humidity	Line trend	% RH; mark missing periods.
CO_2	Line trend	ppm; show timestamp; avoid unsupported threshold claims.
HVAC	Status badge / state timeline	Active, Idle, OFF; never present as a control switch.
Lighting	Status badge / state timeline	ON, OFF, Dimmed; never present as a control switch.
Waste alerts	Alert cards/table + room marker	Severity, duration, issue, wasted kWh if available.
Room utilization	Heatmap where supported	Clearly label time and utilization scale; do not imply unsupported forecasting.

Charts should prioritize honest scales, explicit units, readable legends and exact tooltips. If data is incomplete, the chart should show the gap or an explicit incomplete-data message rather than connecting unsupported values.

29. UI IMPLEMENTATION GUIDELINES

29.1 React mapping

UI layer	Implementation guidance
App shell	Shared layout with header/sidebar and role-aware navigation.
Page components	Dashboard, BuildingView, RoomDetail, Alerts, Recommendations, Analytics, Login.
Reusable components	MetricCard, StatusBadge, RoomNode, AlertRow/Card, RecommendationCard, ChartContainer, EmptyState, ErrorState, Skeleton.
State	Use React/local state for UI concerns; a dedicated state library is not required by the architecture.
Data access	Consume backend/API services; do not connect the browser directly to PostgreSQL.
Role rendering	Use authenticated role context to control visible actions; backend remains authoritative.
Spatial renderer	Implement SVG/JSON floor layouts for the current source-supported MVP; keep the renderer isolated so 3D can be added if confirmed.
Data states	Represent loading, success, empty, partial/stale and error as explicit UI states.
Performance	Avoid unnecessary repeated room queries; use focused API responses and aggregated history for charts.

29.2 Technical boundary

The SAD defines React.js as confirmed frontend technology, FastAPI as the backend, PostgreSQL as persistence, and REST-style application APIs with exact endpoint names still Proposed. ESP32/MQTT is optional for selected real telemetry. The frontend should consume data through the backend/API architecture.

29.3 What the UI must not implement

- Direct HVAC/lighting/breaker control.
- Individual identity tracking, facial recognition or biometrics.
- Native mobile application behavior.
- VR/AR walkthroughs.
- Unapproved advanced predictive maintenance or forecasting.
- Unapproved 3D/WebGL as a finalized MVP requirement.

30. WIREFRAME-LEVEL SCREEN SPECIFICATIONS

30.1 Login Page

Purpose: authenticate users before protected features. Layout: branded header + centered form card. Main content: identifier, password, sign-in, validation/error region. Actions: submit. Loading: button progress. Empty: required-field messages. Error: generic authentication failure. Responsive: centered card with readable width. Accessibility: labels, focus order, live error region.

30.2 Main Dashboard

Purpose: high-level operational overview. Header: current role/user, refresh/freshness. Sidebar: primary navigation. Main: four KPI cards → spatial view + active alerts → compact analytics. Actions: open room, open alert, navigate analytics. Loading: skeleton cards/map/list. Empty: per-widget no-data. Error: localized retry. Responsive: KPI wrap, map remains primary.

30.3 Digital Twin / Building View

Purpose: spatial navigation. Header: building selector + floor selector. Main: map/canvas + legend + selected room context. Actions: select building/floor/room. Loading: map skeleton. Empty: no configured floor. Error: map-load retry. 2D is directly supported by PRD/SRS; 3D renderer remains TBD.

30.4 Room Detail

Purpose: inspect a selected room. Header: Room ID + occupancy status + freshness. Main: occupancy card → environment cards → energy/resource cards → 24-hour preview → alerts/recommendations. Role access follows SRS. Unavailable metrics remain visibly unavailable.

30.5 Alerts Page

Purpose: review energy-waste events. Header: title + filters. Main: table/card list with room, issue, trigger time, duration, severity, status and wasted kWh. Actions: filter/open/acknowledge/resolve only when permitted. Empty: no active alerts. Error: retry.

30.6 Recommendations Page

Purpose: review human decision-support suggestions. Each card: issue + room → evidence → suggested action → severity/timestamp/status. No card should resemble a hardware control button. Any action is an operational instruction for a person, not a command sent to infrastructure.

30.7 Historical Analytics

Purpose: compare occupancy with energy and supported environmental history. Controls: 24h/7d/30d and building/floor/room filters. Main: occupancy-energy time series, optional environment trend and utilization heatmap. Loading: chart skeleton. Incomplete data: visible gaps/label. Query error: retry.

30.8 Profile / Logout

Purpose: session utility. Show current role/context and logout. Do not introduce a separate rich profile-management feature unless approved by the SRS/implementation plan.

31. SCREEN-TO-REQUIREMENT TRACEABILITY

Screen	Related SRS requirements	Related PRD feature	Primary roles
Login	FR-01–04; AC-01–02	Authentication / user access	All
Dashboard	FR-17–25, FR-26+; AC-12–13	Administrator Dashboard; energy insights	Admin, Energy, Operations
Building View	FR-05–09; AC-03–05	Digital Twin Building Visualization	All
Floor Navigation	FR-06–07; AC-04	Floor navigation	All
Room Detail	FR-08, FR-10–20; AC-05–11	Clickable room metadata; occupancy/environment/energy/resources	All with role limits
Alerts	FR-22–27; AC-12–13	Energy-waste alerts	Admin, Energy, Operations
Recommendations	FR-36–37; AC-14	Decision-support recommendations	Admin, Energy, Operations
Historical Analytics	FR-12, FR-16, FR-24/29 as applicable; AC-15	Historical occupancy vs energy; utilization heatmaps	Admin, Energy; Operations proposed
Role-aware navigation	FR-03–04; AC-02	RBAC	All

31.1 Source alignment notes

- SRS §16 defines the eight core screens and their loading/empty/error behavior.
- SRS §3 defines the role permission matrix; this document preserves Proposed entries as Proposed.
- SRS Appendix B defines 89 mapped rooms and states that ranges are supplied room mappings with no additional rooms.
- PRD §5–7 defines the 2D floor map MVP, room status overlays, waste detection, dashboard metrics and historical analytics.
- SAD §6/§10/§14/§19–21 defines the React/API/RBAC/performance and representation-neutral implementation boundaries.
- Phase-I Report pp. 4–8 describes the 3D Digital Twin concept and indoor environmental monitoring; it is treated as concept/future-direction input rather than a silently confirmed MVP renderer.

32. UI/UX ACCEPTANCE CHECKLIST

☐ Login form supports valid authentication and generic invalid-credential feedback.

☐ Logout terminates the authenticated session; protected pages require authentication.

☐ Role-based UI visibility follows the SRS permission matrix.

☐ SVH, Engineering Block and Law Block are selectable.

☐ Configured floors load only their supplied room IDs.

☐ Room ranges are treated as inclusive; total current MVP room count is 89.

☐ Room selection opens a detail view with all available valid telemetry.

☐ Occupancy status and count are visible when available, with timestamp.

☐ Temperature, humidity and CO_2 are visible when valid telemetry exists.

☐ Energy is shown with correct kW/kWh units.

☐ HVAC and lighting states are visible and are not rendered as physical-control switches.

☐ Waste alerts show room, issue/type, duration/timestamp, severity/status and available wasted kWh.

☐ Recommendations show issue, room, evidence, suggested action, severity, timestamp and status.

☐ Historical analytics support 24h, 7d and 30d views.

☐ Loading states exist for all major screens/widgets.

☐ Empty states exist for no alerts, no recommendations, no telemetry and no history.
☐ Error states provide actionable retry/navigation feedback without stack traces.
☐ Stale/offline/partial telemetry is visible and never converted to zero.
☐ Desktop layouts support 1920×1080 and 1366×768.
☐ Keyboard navigation and visible focus states work.
☐ Status is not communicated by color alone.
☐ No PII or individual identity tracking is displayed.
☐ No autonomous physical-control functionality is represented.
☐ 2D/3D spatial direction remains explicitly TBD until confirmed.
☐ Future features are not presented as MVP requirements.
☐ Frontend consumes backend/API services rather than connecting directly to PostgreSQL.

33. UI/UX DESIGN DECISIONS

Decision	Status	Reason
React-based dashboard	Confirmed	SAD/SRS define React.js as the frontend implementation.
Desktop-first responsive layout	Confirmed	SRS prioritizes 1920×1080 and 1366×768 desktop layouts.
Three-building MVP navigation	Confirmed	Current SRS baseline explicitly maps SVH, Engineering Block and Law Block.
89-room inclusive mapping	Confirmed	SRS Appendix B derives 89 rooms directly from supplied ranges; no additional rooms are defined.
2D spatial MVP support	Source-supported	PRD/SRS support lightweight interactive 2D SVG/JSON floor layouts.
3D Digital Twin renderer	TBD	Phase-I describes 3D, while PRD excludes complex 3D in MVP; SRS/SAD preserve this conflict.
Specific icon library	Proposed	Source documents do not finalize an icon library.
Specific chart library	Proposed	Source documents define chart types/needs, not a library.
Native mobile app	Out of scope	SRS/PRD explicitly define web-only MVP.
Physical control UI	Out of scope	System is monitoring and decision support only.
Direct database access from UI	Not allowed by architecture	Frontend communicates through backend/API boundary.

34. FINAL UI/UX SUMMARY

CampusTwin EnergyMind should feel like a focused facilities-monitoring workspace rather than a generic analytics portal. The user enters through authenticated, role-aware access, reaches a high-level dashboard, selects a building and floor, inspects rooms, follows evidence into alerts and recommendations, and uses historical analytics to understand occupancy-energy relationships.

The experience is intentionally structured as:

↓ Login

↓ Role-based Dashboard

↓ Building Selection

↓ Floor Selection

↓ Digital Twin / Spatial View

↓ Room Details

↓ Alerts / Recommendations

↓ Historical Analytics

The core interaction pattern is Awareness → Investigation → Evidence → Decision Support. The system helps authorized people understand building conditions and prioritize inspection; it does not directly control HVAC, lighting, breakers or other physical infrastructure.

FINAL CONSISTENCY RESULT This UI/UX baseline is aligned to the uploaded PRD, SRS, SAD and Phase-I report. Confirmed requirements remain confirmed, Proposed choices remain Proposed, the 2D/3D representation conflict remains TBD, and future features are not presented as current MVP behavior.

Appendix A — Current MVP Room Inventory

Building	Floor	Count	Exact inclusive range	Expanded interpretation
SVH	Ground	12	101–112	101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112
SVH	Second	13	201–213	201 through 213 inclusive
Engineering Block	Ground	8	101–108	101 through 108 inclusive
Engineering Block	First	8	201–208	201 through 208 inclusive
Engineering Block	Second	8	301–308	301 through 308 inclusive
Engineering Block	Third	8	401–408	401 through 408 inclusive
Law Block	Ground	8	101–108	101 through 108 inclusive
Law Block	First	8	201–208	201 through 208 inclusive
Law Block	Second	8	301–308	301 through 308 inclusive
Law Block	Third	8	401–408	401 through 408 inclusive
Total	Current MVP	89	—	No additional rooms defined

Appendix B — Source Reference Map

Source	Relevant sections/pages used
SRS v1.0	§2.7 MVP scope; §3 roles/permissions; §5 functional requirements; §9 authentication; §10 NFRs; §14 error handling; §15 recommendations; §16 UI requirements; §17 acceptance; Appendix B room inventory.
SAD v1.0	§4 technology stack; §6 frontend architecture; §10 API architecture; §14 authentication/RBAC; §15 security; §17 deployment; §19 performance; §20 resilience; §21 decisions.
PRD	§3 target users; §4 product goals/non-goals; §5 core features; §6 MVP scope; §7 user stories; §8 functional requirements.
Phase-I Report	Abstract and pp. 4–8 covering 3D Digital Twin concept, environmental parameters, IoT/ESP32/MQTT and occupancy-related analysis.

Appendix C — Implementation Handoff Notes

- Build the shared application shell first: Header + Sidebar + Breadcrumb + role context.
- Implement building/floor/room navigation against configuration-driven data so the 89-room MVP does not require hard-coded page copies.
- Create a reusable RoomNode that accepts room ID, occupancy state, waste state, freshness and selection state.
- Create a reusable MetricCard that supports valid, unavailable and stale values.
- Create a reusable AlertCard/Row and RecommendationCard with evidence fields.

- Create a reusable AsyncState wrapper pattern for loading, empty, partial and error conditions.
- Keep the spatial renderer behind a stable interface so SVG/JSON and a future 3D renderer can share the same navigation and room-detail contracts.
- Keep chart data aggregated through backend APIs and render only the required time series in the browser.
- Treat backend RBAC as authoritative even when the frontend hides unauthorized controls.